

Power Terminal

Infrastructure: Power Grid

Budget: 1100

Location: Dubai, UAE

Target Year: 2040

Risk Level: LOW

Risk Score: 0.41/100

Climate Resilience and Engineering Report for Dubai Power Grid

The report is based on the provided ground truth data and focuses on enhancing the resilience of the power grid infrastructure in Dubai against flood hazards.

STRATEGY 1

Implementation of Elevated Equipment Platforms

Elevating critical power grid equipment such as transformers and switchgear on platforms higher than the projected flood levels will protect them from inundation. This strategy involves conducting a thorough site analysis to determine the optimal platform height, considering the projected sea level rise and extreme rainfall events. By doing so, the power grid's operational integrity can be maintained during flood events.

STRATEGY 2

Flood-Resistant Cable Systems and Substations

Upgrading the power grid's cable systems and substations with flood-resistant materials and designs will enhance their ability to withstand floodwaters. This includes using waterproof cables, sealable substations, and applying specialized coatings to protect equipment from corrosion. Such measures will minimize the risk of power outages and equipment damage during flood events.

STRATEGY 3

Smart Grid Integration with Flood Monitoring Systems

Integrating the power grid with advanced flood monitoring systems will enable real-time monitoring of flood levels and prediction of potential flood events. This integration allows for proactive measures such as load shedding, grid segmentation, and targeted maintenance to ensure grid resilience. By leveraging smart grid technologies and flood monitoring data, the power grid can be operated more efficiently and safely during flood events.

RESILIENCE SCORE: 87%

